

The First Workshop on Multiple Input Modalities and Sensations for VR/AR Interactions (MIMSVAI)

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ABSTRACT

With the advance of VR/AR technology, more and more VR/AR applications are emerging and have been popular among new users. Users interact with both physical or virtual worlds through multiple input modalities and sensations to create an immersive VR/AR experience. However, a lack of robust and intuitive interaction interfaces and realistic sensations hinders users' experience for achieving a fascinating acceptance in various application areas of VR/AR interactions. This workshop discusses the challenges and applications of designing a higher coherence between different input modalities and sensations to offer more engaging VR/AR experiences, which can create opportunities for the researchers from both UbiComp and VR/AR fields to jointly discuss and brainstorm new directories of designing new input modalities and sensations for VR/AR interactions.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Virtual reality; Augmented reality; Multiple sensory modalities; Multi-sensations; Mixed reality; Realism; User friendliness

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1 OBJECTIVE

With the advance of Virtual Reality and/or Augmented Reality (VR/AR) technology, more and more VR and AR apps are emerging and have been really very popular among new users [1]. To facilitate user interaction with the systems, multiple user input modalities are required. Furthermore, coherent integration of multiple realistic sensations can increase the level of realism [2], and therefore creates a truly immersive VR/AR experience. However, a lack of robust and intuitive interaction interfaces and realistic sensations hinders users' experience for achieving a fascinating acceptance in various application areas of VR/AR interactions. Therefore, many researchers are devoted their efforts to study how to achieve a higher level of coherence between different input modalities and sensations to provide more engaging VR/AR experiences.

Existing studies incorporate a wide range of input modalities (e.g., eye gaze [5], gesture [10], and voice [7]) to sense users' actions so that they can interact with virtual/augmented content as easily as they interact with the real world around them. Besides providing easy-to-use input modalities, many researchers explored ways to enhance realism in the VR/AR world to improve the level of immersiveness. Various sensations (e.g., visual perception[3], auditory cues [4], haptic feelings [9], and olfactory sensations [6]) were incorporated into different VR/AR applications. Leveraging the advances of a combination of VR/AR software and hardware, VR/AR user experience has greatly improved recently, which attracts an increasing number of interdisciplinary researchers, designers, and engineers to apply VR/AR technology into various application domains. However, due to insufficient support of input modalities or sensations, current VR/AR interactions are still not able to provide users with realistic and immersive feelings. To continue improving VR/AR user experience, this workshop explores to bring in more talented UbiComp researchers on studying how to design better input modalities and sensations.

In order to gather research momentum from the UbiComp community, we call for papers discussing how to design or incorporate different input modalities or sensations for enabling a more immersive VR/AR experience. We arrange sessions for each accepted paper to be presented in the workshop. At the end of each session, we will assign session chairs to mediate the discussion to distill motivated ideas and findings to the participants. Besides the regular sessions, we will invite two experts for VR/AR interactions to give talks in the morning and the afternoon, respectively. We

hope these two keynote presentations can bring in more interesting perspectives regarding VR/AR-related researches. Finally, a panel discussion will be held in the late afternoon. We will invite 4 panelists from both academia and industrial sides to share their valuable experiences related to current research or development trends of VR/AR applications.

2 RELEVANT AND TIMELY

Ubiquitous Computing (UbiComp) has always been a field that focuses on bringing computing to people's physical world. Mark Weiser stated that "UbiComp is roughly the opposite of virtual reality. Where virtual reality puts people inside a computer-generated world, ubiquitous computing forces the computer to live out here in the world with people" [8]. On the contrary, Virtual Reality (VR) simulates the physical world and modifies human sensory input to put people in the virtual computing world. In between these two extremes, augmented reality (AR) plays an intermediate role to enrich, rather than replace the physical world with mobile or wearable devices.

As the technology advances in the UbiComp or VR/AR field, new possibilities of applying the combinations of these technologies in different domains emerges, which blurs the line between UbiComp and VR/AR technologies. VR/AR is melding with UbiComp field. UbiComp technology interacts with users through multiple sensory and feedback technologies in the physical world through mobile and wearable devices. The same sets of sensory and feedback technologies can be used to design multiple input modalities and sensations for users to immersively interact with both physical and virtual worlds through VR/AR devices. Moreover, both UbiComp and VR/AR fields require Human-Computer Interaction (HCI) methodologies to incorporate the designed solutions into different application domains to increase their impact. Therefore, we expect this workshop can create opportunities for the researchers from both UbiComp and VR/AR fields to jointly discuss and brainstorm new directories of designing new input modalities and sensations for VR/AR interactions.

3 TOPICS

The workshop fosters discussions among UbiComp and VR/AR fields to cover (1) better input modalities, (2) more realistic sensations, (3) alternative devices and infrastructures, (4) novel application scenarios, and (5) better design process, platform/tools, or libraries for enabling immersive VR/AR interactions. We expect contributions (but not limited to) on the following topics:

- **Better input modalities.** To sense users' actions/intentions, what are the suitable input modalities to be incorporated in to interact with the virtual world? Or how to improve existing or combine multiple input modalities to provide more engaging VR/AR interactions?
- **Realistic human sensations.** To immersively feel the response/change of the virtual, what are the better sensations that can improve the level of realism of the virtual world? Or how to combine or improve existing sensations to interact with the virtual world?
- **Alternative devices and infrastructures.** To enable VR/AR interactions, what are the alternative VR/AR devices and

infrastructures? How to integrate the designed input modalities or sensations with the VR/AR devices or infrastructures?

- **Novel application scenarios.** VR/AR interactions have been used in many application domains, e.g., gaming, health-care, or education. Gartner expects it will take 5 to 10 years before immersive technologies such as VR and AR reach a mature level. As the VR/AR technology continues to mature, which applications are the most useful for using the future VR/AR interactions that can create the most impact on society. What are the best application domains to apply VR/AR interactions that can have the strongest research and/or industrial impact?
- **Suitable design process, platform/tools, or libraries.** To design or prototype VR/AR applications, what are the better design processes, platforms/tools (e.g., Unity), or libraries (e.g., Android ARCore or Apple ARKit) that should be proposed or designed?

4 WORKSHOP STRUCTURE

The workshop will follow a full single-day format. We plan different activities during the workshop to make it more interactive, including the introduction to the workshop, two keynote talks, paper presentation, panel session, and Best Paper Award (see Table 1). Particularly, we will invite two experts for VR/AR interactions to give talks so the workshop can bring in more interesting perspectives regarding VR/AR-related researches in the UbiComp community. In the panel session, we will invite 4 panelists from both academia and industrial sides to share their valuable experiences related to current research or development trends. Participants can also discuss topics that they are interested in with the panelists.

Table 1: Workshop Schedule

Workshop Schedule	
9:00-9:10 am	Opening
9:10am-10:20am	Keynote Talk I: 50 + 20min
10:20am-10:30am	Coffee Break
10:30am-11:10am	Paper Session I: $2 \times (15 + 5)$ min
11:10am-11:30am	Coffee Break
11:30am-12:30pm	Paper Session II: $3 \times (15 + 5)$ min
12:30pm-2:00pm	Lunch Break
2:00pm-3:10pm	Keynote Talk II: 50 + 20min
3:10pm-3:20pm	Coffee Break
3:20pm-4:00pm	Paper Session III: $2 \times (15 + 5)$ min
4:00pm-4:20pm	Coffee Break
4:20pm-5:20pm	Panel Session 60min
5:20pm-5:30pm	Best Paper Award

The papers need to be with a maximum length of 5 pages (4-page + 1 references) in ACM SIGCHI Master Article template. The accepted papers will be published in the UbiComp/ISWC Adjunct Proceedings, which will be included in the ACM Digital Library. The Best Paper Award is given to the best paper presented at the MIMSVAI 2021 workshop, to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

The call for papers will be sent to all relevant mailing lists of several research communities, including those of UbiComp research, HCI, VR/AR applications, haptic and tactile interfaces, and multi-sensory perception communities.

At least two reviewers from our organizers and PC members will peer-review each submission. Our acceptance criteria will be a mixture of novelty, research quality, and relevance. We expect to select 6 accepted papers. Given the theme of this workshop and the rapid growth of the VR/AR community, we expect to attract a large number of submissions which enables us to organize a high-quality workshop.

5 EXPECTED OUTCOMES

The workshop will be open both to the attendees with an accepted paper and to researchers simply interested in VR/AR interactions. We expect to attract 15-20 participants. The workshop will lead to a deeper understanding of the multiple input modalities and sensations for VR/AR interactions and provide knowledge about the current state-of-the-art studies around the workshop scheme.

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REFERENCES

- [1] 2019. The Real Significance Of AR, VR And Mixed Reality. <https://www.forbes.com/sites/timbajarin/2019/09/23/the-real-significance-of-ar-vr-and-mixed-reality/?sh=2f06b2cf6d80>.
- [2] Alan Chalmers and Andrej Ferko. 2008. Levels of Realism: From Virtual Reality to Real Virtuality. In *Proceedings of the 24th Spring Conference on Computer Graphics* (Budmerice, Slovakia) (SCCG '08). Association for Computing Machinery, New York, NY, USA, 19–25. <https://doi.org/10.1145/1921264.1921272>
- [3] Valeria Garro, Veronica Sundstedt, and Diego Navarro. 2020. A Review of Current Trends on Visual Perception Studies in Virtual and Augmented Reality. In *SIGGRAPH Asia 2020 Courses* (Virtual Event) (SA '20). Association for Computing Machinery, New York, NY, USA, Article 7, 97 pages. <https://doi.org/10.1145/3415263.3419144>
- [4] Mar Gonzalez-Franco, Antonella Maselli, Dinei Florencio, Nikolai Smolyanskiy, and Zhengyou Zhang. 2017. Concurrent talking in immersive virtual reality: on the dominance of visual speech cues. *Scientific Reports* 7, 1 (19 Jun 2017), 3817. <https://doi.org/10.1038/s41598-017-04201-x>
- [5] Hae-Jong Joo and Hwa-Young Jeong. 2020. A study on eye-tracking-based Interface for VR/AR education platform. *Multimedia Tools and Applications* 79, 23 (01 Jun 2020), 16719–16730. <https://doi.org/10.1007/s11042-019-08327-0>
- [6] Erika Kerruish. 2019. Arranging sensations: smell and taste in augmented and virtual reality. *The Senses and Society* 14, 1 (2019), 31–45. <https://doi.org/10.1080/17458927.2018.1556952> arXiv:<https://doi.org/10.1080/17458927.2018.1556952>
- [7] T. Piumsomboon, D. Altimira, H. Kim, A. Clark, G. Lee, and M. Billinghurst. 2014. Grasp-Shell vs gesture-speech: A comparison of direct and indirect natural interaction techniques in augmented reality. In *2014 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. 73–82. <https://doi.org/10.1109/ISMAR.2014.6948411>
- [8] Mark Weiser. 1991. The Computer for the Twenty-First Century. *Scientific American* 265, 3 (1991), 94–110.
- [9] Eric Whitmire, Hrvoje Benko, Christian Holz, Eyal Ofek, and Mike Sinclair. 2018. Haptic Revolver: Touch, Shear, Texture, and Shape Rendering on a Reconfigurable Virtual Reality Controller. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). ACM, New York, NY, USA, Article 86, 12 pages. <https://doi.org/10.1145/3173574.3173660>
- [10] Yukang Yan, Yingtian Shi, Chun Yu, and Yuanchun Shi. 2020. HeadCross: Exploring Head-Based Crossing Selection on Head-Mounted Displays. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 4, 1, Article 35 (March 2020), 22 pages. <https://doi.org/10.1145/3380983>

A APPENDIX: CALL FOR PAPERS

With the advance of VR/AR technology, more and more VR/AR applications are emerging and have been popular among new users. To facilitate user interaction with the systems, multi-modal input modalities are required. Furthermore, coherent integration of multiple realistic sensations can increase the level of realism, and therefore creates a truly immersive VR/AR experience. However, a lack of robust and intuitive interaction interfaces and realistic sensations hinders users' experience for achieving a fascinating acceptance in various application areas of VR/AR interactions. This workshop discusses the challenges and applications of designing a higher coherence between the different input modalities and sensations to provide more engaging VR/AR experiences by leveraging the advances of both VR/AR software and hardware to improve user experience.

Papers may include, but not be limited to, topics such as:

- 2D/3D and volumetric display and projection technology
- Immersive analytics and visualization
- Modeling and simulation
- Multimodal capturing and reconstruction
- Scene description and management issues
- Storytelling
- Tracking and sensing
- Audio interfaces, sound rendering, spatialized audio, auditory perception and psychoacoustics
- Embodied agents, virtual humans and (self-)avatars
- Haptic and tactile interfaces, wearable haptics, passive haptics, pseudo haptics, other touch-based UI
- Mediated and diminished reality
- Multimodal input and output
- Multisensory rendering, registration, and synchronization
- Perception and cognition
- Presence, body ownership, and agency
- Teleoperation and telepresence
- 3D user interaction
- 3D UI metaphors
- Collaborative interactions
- Ethical issues
- Human factors and ergonomics
- Input devices
- Locomotion and navigation
- Multimodal/cross-modal interaction and perception
- Non-fatiguing 3D UIs
- Non-visual interfaces (such as olfactory)
- Touch, tangible and gesture interfaces
- Usage research, evaluation methods and empirical studies
- Domain-specific VR/AR applications.
- Design process, platform/tools, or libraries used to incorporate input modalities or sensations for VR/AR interactions or studying their use

Important Dates

Paper submission: June 30, 2021

Paper notification: July 15, 2021

Camera-ready: July 31, 2021

Workshop: September 26, 2021

Best Paper Award

The Best Paper Award is given to the best paper presented at the MIMSVAI 2021 workshop, to acknowledge and encourage excellence in research. The awardee will be presented with a certificate at the workshop.

Workshop Organizer

Chuang-Wen You, National Tsing Hua University

Yi-Chao Chen, Shanghai Jiao Tong University

Hsin-Ruey Tsai, National Chengchi University

Bin Sheng, Shanghai Jiao Tong University